

DUKAN: Disentangled and Uncertainty-Aware Kolmogorov–Arnold Network for Multimodal Sentiment Analysis

Anonymous Author(s)

Abstract. Multimodal Sentiment Analysis (MSA) aims to integrate complementary cues from language, acoustic, and visual channels, yet two obstacles persist in MSA pipelines. First, each modality comprises a shared sentiment consensus and modality-unique cues, yet a common practice is to encode each modality independently and leave the two entangled in a single latent space. Second, modality reliability varies sharply across samples, whereas conventional fusion applies one averaged weighting profile that tends to let the information-dense linguistic channel dominate the weaker ones. To tackle both issues, we introduce DUKAN, a disentangled and uncertainty-aware framework built on an interpretable Kolmogorov–Arnold Network (KAN) fusion backbone. Its cross-modal disentangled information bottleneck (CM-DRD-MIB) factorizes each modality into a shared and a specific subspace via an InfoNCE alignment loss across modalities and a redundancy-reduction disentanglement loss within each modality. A lightweight gate (UAMG) reuses each per-modality encoder’s shared and specific posterior standard deviations as a per-sample uncertainty signal and converts them into sample-level modality weights through a temperature-controlled softmax. Both modules are trained under Pareto-coordinated unimodal and multimodal objectives. On CMU-MOSI, CMU-MOSEI, and the Chinese CH-SIMS v2, DUKAN attains state-of-the-art or competitive performance while improving robustness to modality corruption. Our code will be made publicly available.

Keywords: Multimodal Sentiment Analysis · Kolmogorov–Arnold Networks · Information Bottleneck · Disentangled Representation · Uncertainty-Aware Fusion

1 Introduction

Multimodal Sentiment Analysis (MSA) infers affective states from complementary signals such as language, acoustic prosody, and facial expression. Since sentiment is often ambiguous within any single channel, modeling cross-modal interactions is essential [1,2]. Two key challenges remain. First, the modalities carry overlapping sentiment cues, yet each also carries cues the others cannot recover; most pipelines encode each modality independently, redundantly representing the same content while entangling modality-unique cues with that

redundancy. Second, a modality’s usefulness varies sharply across samples: audio is silent in quiet segments, vision is uninformative when faces are occluded, and text is misleading under sarcasm. Yet most fusion modules learn a single modality-weighting profile averaged over training, letting the dominant modality (typically text) drown out the others where it should not.

Recent studies partially mitigate these limitations. MMIM [3] maximizes mutual information between modalities but without explicit subspace structure, so shared and modality-unique components remain entangled. ITHP [4] compresses cross-modal redundancy via an information-theoretic primer, yet exploits no per-sample encoder uncertainty for reweighting and relies on a fixed fusion path. KAN-MCP [5] purifies each modality through variational information-bottleneck encoders and shows that interpretable KAN-based fusion is feasible. However, its encoders run independently per modality without cross-modal alignment, and the per-sample standard deviations estimated inside each per-modality IB encoder are discarded after reparameterization rather than reused for gating. Explicit cross-modal structure and sample-level modality reliability thus remain insufficiently modeled.

To address both issues, we propose **DUKAN**, a disentangled and uncertainty-aware framework built on an interpretable KAN backbone. Our starting point is an observation prior IB-based pipelines overlook: a variational IB encoder already produces two by-products it normally discards—a latent posterior separable into shared and specific parts, and a per-sample posterior standard deviation measuring each modality’s reliability on each input. DUKAN turns these existing by-products into two lightweight modules, plus a gradient-level balancer, improving the representations the KAN consumes rather than enlarging the fusion operator.

Concretely, the *Cross-Modal DRD-MIB* (CM-DRD-MIB) encoder—a cross-modal extension of the per-modality DRD-MIB (Dimensionality-Reduction and Denoising Modal Information Bottleneck) in KAN-MCP [5]—compresses each modality under the IB objective and factorizes its posterior into a shared and a specific subspace, aligning the shared subspaces across modalities with an InfoNCE loss and decorrelating shared from specific within each modality with a Barlow-Twins penalty, so the KAN receives clean, low-dimensional, interpretable channels. The *Uncertainty-Aware Modality Gating* (UAMG) module then recycles the per-modality posterior standard deviations as a per-sample uncertainty signal, softmax-mapped to modality weights that down-weight a modality exactly when its encoder is uncertain—without a dedicated uncertainty head. Because the linguistic gradient can still overwhelm the weaker streams during joint training, an MMPareto strategy coordinates the multimodal and unimodal objectives per modality, giving a two-level, sample- and gradient-level balancing mechanism. The main contributions of this work are summarized as follows:

1. We propose **CM-DRD-MIB**, which factorizes each modality into an InfoNCE-aligned shared subspace and a Barlow-Twins-decorrelated specific subspace, handing the KAN clean, disentangled channels.

2. We propose **UAMG**, a lightweight gate that turns each per-modality encoder’s shared and specific posterior standard deviations into sample-level modality weights through a temperature-controlled softmax, with no dedicated uncertainty head.
3. Extensive experiments on CMU-MOSI [6], CMU-MOSEI [1], and the Chinese CH-SIMS v2 [7] show that DUKAN achieves state-of-the-art or competitive performance (Acc-2 89.62, F1 89.61 on CMU-MOSI; Acc-7 54.68 on CMU-MOSEI) with improved robustness to modality corruption.

2 Related Work

Multimodal Sentiment Analysis and Interpretable Fusion. MSA research spans two broad families. Early methods design explicit fusion operators, e.g., the outer-product tensor fusion of TFN [8] and the cross-modal attention of the Multimodal Transformer [2]. More recent work imposes representation-level objectives, such as mutual information maximization (MMIM [3]), modality-invariant/specific decomposition (MISA [9]), contrastive feature decomposition (ConFEDE [10]), and IB-based compression (ITHP [4]). These mechanisms are largely opaque. KAN-MCP [5] instead adopts Kolmogorov–Arnold Networks (KAN [11]), which place learnable B-spline activations on edges to realize the Kolmogorov–Arnold representation theorem [12], decomposing fusion into inspectable univariate functions. In this work, we adopt KAN-MCP as our backbone and enhance its input representations rather than improving the fusion operator.

Information-Theoretic and Disentangled Representations. Our representation design builds on three threads: the IB principle [13], made tractable by the deep variational IB [14] and extended to multimodal settings [15]; InfoNCE [16], which aligns representations while lower-bounding mutual information; and Barlow Twins [17], which reduces redundancy by driving the cross-correlation toward the identity. Our CM-DRD-MIB combines all three: a decomposition of each per-modality IB encoder into shared and specific subspaces, an InfoNCE term that aligns shared subspaces, and a Barlow-style independence penalty between them.

Balanced and Uncertainty-Aware Multimodal Learning. Jointly trained multimodal models tend to over-rely on dominant modalities [18]. This motivates gradient-level rebalancing, e.g., on-the-fly gradient modulation [19], classifier-guided modulation [20], and Pareto-based coordination (MMPareto [21], adopted by the baseline [5]). Such methods act at the global, batch-level gradient scale. A complementary line instead weights modalities by per-sample reliability: Trusted Multi-View Classification [22] fuses at the evidence level, and Quality-aware Multimodal Fusion [23] weights each modality inversely to its uncertainty, both typically requiring dedicated uncertainty heads. Our UAMG instead reuses the diagonal posterior standard deviations already produced by each per-modality IB encoder as a per-sample uncertainty signal and applies a detached softmax gate, combining sample-level gating with global MMPareto coordination into a two-level balancing mechanism.

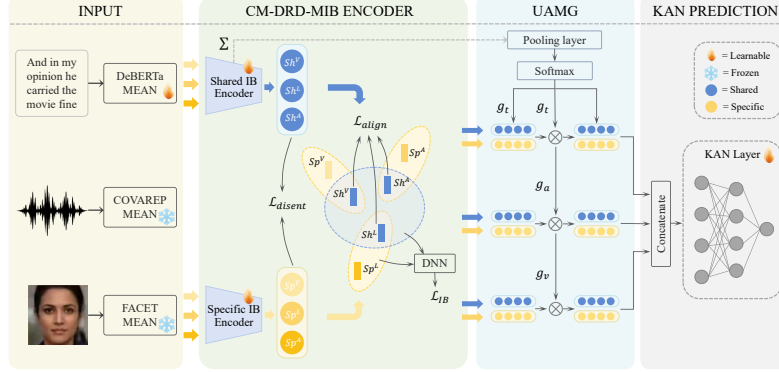


Fig. 1: Overall framework of the proposed DUKAN model.

3 Method

3.1 Overview

A KAN backbone makes fusion inspectable through its edge-level univariate splines, but it degrades on high-dimensional, noisy multimodal vectors and over-relies on the information-dense linguistic modality—an imbalance the dominant linguistic gradient further reinforces during optimization. As outlined in Section 1 and Figure 1, DUKAN therefore (i) extracts sentence-level unimodal features, (ii) passes each modality through a *Cross-Modal DRD-MIB* (CM-DRD-MIB) encoder that compresses it and disentangles a shared (cross-modal consensus) from a specific (modality-unique) subspace, (iii) reweights the modalities per sample via UAMG from the posterior standard deviations the encoder already produces, and (iv) fuses the gated representations with a KAN, all coordinated per modality by a Pareto strategy. We detail each block below.

3.2 Unimodal Feature Extraction

For the linguistic modality we adopt the pretrained **DeBERTa-v3-base** encoder as the language backbone, mapping I_t to $x_t \in \mathbb{R}^{T_t \times d_t}$; for the acoustic and visual modalities we use the pre-extracted frame-level features $x_a \in \mathbb{R}^{T_a \times d_a}$ (COVAREP) and $x_v \in \mathbb{R}^{T_v \times d_v}$ (FACET). Here T_m is the number of tokens/frames and d_m the feature dimension of modality m . A temporal average pooling yields a sentence-level vector

$$\bar{x}_m = \frac{1}{T_m} \sum_{f=1}^{T_m} x_m^{(f)} \in \mathbb{R}^{d_m}, \quad m \in \{t, a, v\}, \quad (1)$$

where $x_m^{(f)}$ is the f -th frame feature. These pooled vectors are the input to CM-DRD-MIB.

3.3 Cross-Modal DRD-MIB

Pooled features are ill-suited to a KAN: they are high-dimensional and noisy, which the edge-spline KAN handles poorly, and they entangle cross-modal consensus with modality-unique information, obscuring the interpretability the KAN is meant to provide. Following the variational information-bottleneck (IB) principle, CM-DRD-MIB compresses each modality into a minimal sufficient, denoised representation and factorizes it into a shared and a specific subspace. Two regularizers are employed to achieve the factorization: a cross-modal alignment loss forcing the shared subspaces of different modalities to encode the *same* consensus, and a disentanglement loss preventing the specific subspace from re-encoding what the shared one already captures.

Subspace factorization. For modality m , a backbone $E_m(\cdot)$ first encodes the pooled feature, $h^m = E_m(\bar{x}_m)$. Two Gaussian-parameter heads G_s^m and G_p^m then map h^m to the parameters of two diagonal Gaussians,

$$(\mu_s^m, \sigma_s^m) = G_s^m(h^m), \quad (\mu_p^m, \sigma_p^m) = G_p^m(h^m), \quad (2)$$

where $(\mu_s^m, \sigma_s^m) \in \mathbb{R}^{d_s}$ and $(\mu_p^m, \sigma_p^m) \in \mathbb{R}^{d_p}$ are the means and softplus-positive standard deviations of the shared (d_s) and specific (d_p) subspaces. The reparameterized samples are

$$h_s^m = \mu_s^m + \sigma_s^m \odot \epsilon_s, \quad h_p^m = \mu_p^m + \sigma_p^m \odot \epsilon_p, \quad (3)$$

with $\epsilon_s, \epsilon_p \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and $\odot$ the element-wise product. A decoder D_m produces a unimodal prediction $\hat{y}_m = D_m([h_s^m; h_p^m])$ ($[\cdot; \cdot]$ is concatenation). Each modality is regularized by an IB objective combining a distortion and a compression term,

$$\mathcal{L}_m = \ell_1(\hat{y}_m, y) + \beta(\text{KL}_s^m + \text{KL}_p^m), \quad (4)$$

$$\text{KL}_{\bullet}^m = \frac{1}{2} \sum_{j=1}^{d_{\bullet}} \left[(\mu_{\bullet,j}^m)^2 + (\sigma_{\bullet,j}^m)^2 - 2 \log \sigma_{\bullet,j}^m - 1 \right], \quad \bullet \in \{s, p\}, \quad (5)$$

where y is the ground-truth score, ℓ_1 the MAE distortion, β the compression weight, and $\text{KL}_{\bullet}^m$ the closed-form KL divergence between each per-dimension Gaussian and the standard normal prior. This compresses redundancy while preserving the minimal sufficient information about y , giving the KAN clean, compact channels.

Cross-modal alignment loss. The shared subspace is meaningful only if the modalities agree within it. We therefore align same-sample shared representations with a symmetric InfoNCE objective over a minibatch of B samples. Let $\hat{h}_s^{m,(i)}$ be the ℓ_2 -normalized shared representation of sample i under modality m . With the scaled similarity

$$s_{ij}^{mm'} = \langle \hat{h}_s^{m,(i)}, \hat{h}_s^{m',(j)} \rangle / \tau, \quad (6)$$

where $\langle \cdot, \cdot \rangle$ is the inner product, τ the InfoNCE temperature, and i, j index samples, the pairwise loss between modalities m and m' is

$$\text{NCE}(m, m') = -\frac{1}{2B} \sum_{i=1}^B \left[\log \frac{e^{s_{ii}^{mm'}}}{\sum_{j=1}^B e^{s_{ij}^{mm'}}} + \log \frac{e^{s_{ii}^{mm'}}}{\sum_{j=1}^B e^{s_{ji}^{mm'}}} \right]. \quad (7)$$

Shared representations of the same sample across two modalities form positive pairs, while the remaining in-batch samples serve as negatives. The overall alignment loss averages the three modality pairs,

$$\mathcal{L}_{\text{align}} = \frac{1}{3} [\text{NCE}(t, a) + \text{NCE}(t, v) + \text{NCE}(a, v)]. \quad (8)$$

Shared/specific disentanglement loss. Without an explicit constraint, the specific subspace can copy information in the shared one, double-counting cues in the fusion vector F and defeating both the IB compression and channel-level interpretability. We therefore penalize the batch-level cross-correlation between the two subspaces. Let $Z_s^m \in \mathbb{R}^{B \times d_s}$ and $Z_p^m \in \mathbb{R}^{B \times d_p}$ stack the shared and specific representations of the minibatch, and $\bar{Z}_s^m, \bar{Z}_p^m$ their per-dimension batch-standardized versions (zero mean, unit variance). The cross-correlation matrix and the disentanglement loss are

$$\mathcal{C}^m = \frac{1}{B} (\bar{Z}_s^m)^\top \bar{Z}_p^m \in \mathbb{R}^{d_s \times d_p}, \quad \mathcal{L}_{\text{disent}} = \frac{1}{3} \sum_{m \in \{t, a, v\}} \frac{1}{d_s d_p} \|\mathcal{C}^m\|_F^2, \quad (9)$$

where $\|\cdot\|_F$ is the Frobenius norm. Driving every entry of $\mathcal{C}^m$ toward zero pushes the two subspaces toward statistical independence, yielding a clean division of labor between cross-modal consensus and modality-unique complementarity.

3.4 Uncertainty-Aware Modality Gating

Modality reliability varies strongly across samples (audio, e.g., carries little signal during silent segments), yet fixed fusion treats every modality equally. UAMG reuses the diagonal standard deviations σ_s^m, σ_p^m already estimated inside CM-DRD-MIB as a per-sample uncertainty signal for adaptive reweighting, without a dedicated uncertainty head.

Per-sample uncertainty. For sample i and modality m , we average the posterior standard deviations over all dimensions of *both* the shared and specific subspaces, yielding a single scalar aleatoric uncertainty

$$u_m^{(i)} = \frac{1}{d_s + d_p} \left(\sum_{j=1}^{d_s} \sigma_{s,j}^{m,(i)} + \sum_{j=1}^{d_p} \sigma_{p,j}^{m,(i)} \right), \quad (10)$$

where $\sigma_{s,j}^{m,(i)}$ and $\sigma_{p,j}^{m,(i)}$ are the j -th shared and specific posterior standard deviations of modality m on sample i , and a larger $u_m^{(i)}$ indicates lower confidence in modality m on that sample.

Sample-level softmax gating. Defining modality quality as the negative uncertainty, we obtain the gate of each modality through a temperature-controlled softmax

$$g_m = 3 \cdot \frac{\exp(-\text{sg}[u_m]/T_g)}{\sum_{m' \in \{t,a,v\}} \exp(-\text{sg}[u_{m'}]/T_g)}, \quad (11)$$

where T_g is the gate temperature and $\text{sg}[\cdot]$ the stop-gradient operator. Since a softmax over three modalities has mean $1/3$, the factor 3 restores the mean gate to 1, keeping the KAN inputs at full scale. The same gate scales both subspaces of a modality,

$$\tilde{h}_s^m = g_m \cdot h_s^m, \quad \tilde{h}_p^m = g_m \cdot h_p^m. \quad (12)$$

Forward-only reweighting. The stop-gradient detaches u_m , keeping σ_s^m, σ_p^m a pure uncertainty signal rather than a task-tuned quantity. Gating thus acts as forward-only reweighting, affecting only the fusion input while the unimodal IB objectives use the un-gated representations; our ablation confirms this outperforms an end-to-end-learned counterpart that lets the gradient flow through u_m .

3.5 KAN-based Interpretable Fusion and Prediction

We adopt KAN as the fusion predictor. Unlike MLPs, which place fixed activations on nodes, KAN places learnable univariate splines on the edges,

$$\text{KAN}(F) = (\Phi_{L-1} \circ \Phi_{L-2} \circ \dots \circ \Phi_0)(F), \quad (13)$$

where F is the fusion input defined below, L the number of layers, $\circ$ function composition, and each layer $\Phi_\ell = \{\phi_{q,p}\}$ a matrix of univariate splines on the edges. This structure approximates complex multivariate functions compactly and offers built-in interpretability through edge-level spline visualization. We concatenate the six gated representations as the fusion input and predict

$$F = [\tilde{h}_s^t; \tilde{h}_s^a; \tilde{h}_s^v; \tilde{h}_p^t; \tilde{h}_p^a; \tilde{h}_p^v] \in \mathbb{R}^{3(d_s+d_p)}, \quad \hat{y} = \text{KAN}(F), \quad (14)$$

where $\hat{y}$ is the final prediction (with UAMG disabled, $g_m = 1$). Because the shared and specific channel positions in F are fixed and the disentanglement loss keeps the two groups independent, inspecting the corresponding edge splines reveals how cross-modal consensus and modality-unique cues each drive the decision.

3.6 Training Objective and Gradient Coordination

We cast training as a multi-objective problem. The multimodal objective folds the two regularizers into the task loss,

$$\mathcal{L}_{\text{mm}} = \text{MSE}(\hat{y}, y) + w_{\text{align}} \mathcal{L}_{\text{align}} + w_{\text{disent}} \mathcal{L}_{\text{disent}}, \quad (15)$$

where $\text{MSE}(\cdot, \cdot)$ is the mean-squared error and $w_{\text{align}}, w_{\text{disent}}$ are loss weights. It is complemented by the three unimodal auxiliary objectives $\mathcal{L}_t, \mathcal{L}_a, \mathcal{L}_v$.

Because the linguistic modality is information-dense, its multimodal gradient dominates the shared encoder and can suppress the acoustic and visual auxiliary signals, re-introducing the imbalance CM-DRD-MIB and UAMG remove. We therefore coordinate the multimodal and unimodal objectives *per modality* via a Pareto principle. For modality m with encoder parameters θ_m , define $\nabla_{\text{mm}}^m = \nabla_{\theta_m} \mathcal{L}_{\text{mm}}$ and $\nabla_{\text{uni}}^m = \nabla_{\theta_m} \mathcal{L}_m$. Their cosine similarity is

$$\rho_m = \frac{\langle \nabla_{\text{mm}}^m, \nabla_{\text{uni}}^m \rangle}{\|\nabla_{\text{mm}}^m\|_2 \|\nabla_{\text{uni}}^m\|_2}. \quad (16)$$

The mixing coefficients $\alpha^m = (\alpha_0^m, \alpha_1^m)$ are chosen as

$$\alpha^m = \begin{cases} (\frac{1}{2}, \frac{1}{2}), & \rho_m > 0, \\ \arg \min_{\substack{\alpha_0 + \alpha_1 = 1 \\ \alpha_0, \alpha_1 \geq 0}} \|\alpha_0 \nabla_{\text{mm}}^m + \alpha_1 \nabla_{\text{uni}}^m\|_2^2, & \rho_m \leq 0. \end{cases} \quad (17)$$

When the gradients are non-conflicting ($\rho_m > 0$) they point the same way, so we average them, which also amplifies the SGD noise term and aids generalization; when they conflict ($\rho_m \leq 0$), the min-norm solution gives a conflict-free direction. The coordinated update for θ_m is

$$\Delta_m = \alpha_0^m \nabla_{\text{mm}}^m + \alpha_1^m \nabla_{\text{uni}}^m, \quad (18)$$

which is rescaled to be no smaller than the naive joint gradient and multiplied by an amplification factor $\gamma > 1$ before the optimizer step. Since $\mathcal{L}_{\text{align}}$ and $\mathcal{L}_{\text{disent}}$ are absorbed into $\mathcal{L}_{\text{mm}}$, the same mechanism governs them, leaving w_{align} and w_{disent} as the only manual loss weights.

4 Experimental Methodology

Datasets and Evaluation Metrics. We evaluate on two widely used MSA benchmarks. CMU-MOSI [6] contains 2,199 utterance-level clips (1,281/229/685 train/val/test), scored in $[-3, 3]$; CMU-MOSEI [1] is larger, with 16,265/1,869/4,643 utterances under the same scheme. Following the baseline, text is encoded with DeBERTa [24] (768-d); acoustic/visual features are 74/47-d (CMU-MOSI) and 74/35-d (CMU-MOSEI). To test generality to a Chinese benchmark we also evaluate on the Chinese benchmark CH-SIMS v2 [7], whose stronger non-verbal cues make it a complementary stress test. We report five standard metrics: 7-class accuracy (Acc-7), binary accuracy (Acc-2), F1, mean absolute error (MAE), and Pearson correlation (Corr); higher is better except for MAE. On CH-SIMS v2 we follow the dataset’s 5/3/2-class accuracy protocol.

Implementation Details. Implemented in PyTorch and trained on a single NVIDIA A800, the model retains the KAN fusion predictor and MMPareto coordination of KAN-MCP, replacing DRD-MIB with CM-DRD-MIB and introducing UAMG. On CMU-MOSI we use batch size 8, 35 epochs, learning rates $1\text{e-}5$ (text)/ $1\text{e-}3$ (rest); on CMU-MOSEI, batch size 64, 10 epochs, $1\text{e-}5/1\text{e-}2$,

warmup ratio 0.5. The CM-DRD-MIB intermediate dimension is 1024/518 for CMU-MOSI/CMU-MOSEI, the fusion KAN uses 4 hidden neurons, and the per-modality dimension splits into shared ($sd=2$) and specific ($pd=2$) parts. The InfoNCE temperature is 0.1; the UAMG gate temperature T_g is 2.0 (CMU-MOSI)/1.0 (CMU-MOSEI), with the gate detached from the covariance branch. MMPareto coordinates the objectives (unimodal weight 0.1/1.0, SGD-noise factor $\gamma=1.5$). Unless stated otherwise, analysis experiments use the default ($sd=2$, $pd=2$, $w_{\text{align}}=0.05$, $w_{\text{disent}}=0.05$, $T_g=2.0$) and vary only the studied factor. The headline comparison in Table 1 reports the mean over five random seeds, with hyperparameters selected only on the validation set.

5 Experimental Results

We assess the proposed method from several complementary angles: comparison with state-of-the-art baselines, component and hyperparameter ablations, inference-time robustness under modality corruption, computational cost, and a qualitative analysis of the uncertainty-based gate and the shared/specific representations. Unless noted otherwise, all analysis experiments are carried out on CMU-MOSI.

5.1 Comparison with State-of-the-Art Methods

Table 1 reports the principal numbers against representative baselines; for methods reporting two binary settings we retain the negative/positive values. On CMU-MOSI our full model tops four of the five metrics—Acc-2 (89.62), F1 (89.61), MAE (0.6078) and Corr (0.8587)—with 7-class accuracy the only exception (analyzed below), sharpening binary discrimination while delivering the best regression fit in the table.

On CMU-MOSEI it again tops four of five metrics (Acc-7 54.68, Acc-2 87.21, F1 87.12, MAE 0.5135), second only on Corr; the Acc-7 gain over KAN-MCP (+1.96) and the strongest baseline (+0.78) on the far larger test set shows the modules transfer across scales.

Results on CH-SIMS v2. To confirm the gains are not specific to English corpora, we also evaluate on the Chinese benchmark CH-SIMS v2 (Table 2). DUKAN achieves the best results on five of six metrics and the second-best Acc-5, ahead of the CH-SIMS-tailored AV-MC [7], the knowledge-guided KuDA [32], and the KAN-MCP backbone—indicating that cross-modal disentanglement and uncertainty-aware gating capture language-agnostic structure rather than corpus-specific artifacts.

5.2 Ablation Study

Component ablation. We toggle the three ingredients—cross-modal alignment (w_{align}), disentanglement (w_{disent}), and the gate (UAMG)—one at a time (Table 3). From the bare variant (Acc-2 88.24), disentanglement alone raises Acc-2

Table 1: Comparison with representative baselines on CMU-MOSI and CMU-MOSEI. The best results are marked in **bold**, while the second-best results are underlined. * denotes our reproduction.

Model	CMU-MOSI					CMU-MOSEI				
	Acc-7 $\uparrow$	Acc-2 $\uparrow$	F1 $\uparrow$	MAE $\downarrow$	Corr $\uparrow$	Acc-7 $\uparrow$	Acc-2 $\uparrow$	F1 $\uparrow$	MAE $\downarrow$	Corr $\uparrow$
TFN [8]	34.90	80.08	80.07	0.90	0.70	50.20	82.50	82.10	0.59	0.70
MuT [2]	40.00	83.00	82.00	0.87	0.70	51.80	82.50	82.30	0.58	0.70
Graph-MFN [1]	34.40	80.20	80.10	0.94	0.66	51.90	84.00	83.80	0.57	0.73
MFN [25]	33.30	80.00	80.10	0.95	0.66	50.80	83.40	83.40	0.58	0.72
MISA [9]	41.37	83.54	83.58	0.78	0.78	52.05	84.67	84.66	0.56	0.75
MAG-BERT [26]	43.62	84.43	84.61	0.73	0.78	52.67	84.82	84.71	0.54	0.76
MMIM [3]	45.00	85.10	85.00	0.74	0.78	53.10	85.10	85.00	0.55	0.75
HyCon [27]	46.60	85.20	85.10	0.74	0.78	52.80	85.40	85.60	0.55	0.75
MGCL [28]	49.30	86.70	86.70	0.69	0.71	<u>53.90</u>	86.40	86.40	0.54	0.77
ULMD [29]	47.80	85.80	85.70	0.70	0.80	53.80	86.00	85.90	0.53	0.77
MFON [30]	44.90	86.90	86.90	0.73	0.80	53.70	86.30	86.30	<u>0.53</u>	0.78
DLF [31]	47.08	85.06	85.04	0.73	0.78	53.90	85.42	85.27	0.54	0.76
ITHP [4]	47.70	<u>88.50</u>	<u>88.50</u>	0.66	<u>0.86</u>	52.20	<u>87.10</u>	<u>87.10</u>	0.55	0.79
KAN-MCP* [5]	<u>48.47</u>	88.24	88.20	<u>0.63</u>	0.85	52.72	86.60	86.65	0.54	0.79
Ours	48.32	89.62	89.61	0.61	0.86	54.68	87.21	87.12	0.51	<u>0.79</u>

Table 2: Comparison with representative baselines on CH-SIMS v2. The best results are marked in **bold**, while the second-best results are underlined.

Model	Acc-5 $\uparrow$	Acc-3 $\uparrow$	Acc-2 $\uparrow$	F1 $\uparrow$	MAE $\downarrow$	Corr $\uparrow$
EF-LSTM [33]	53.7	73.5	80.1	80.0	0.309	0.700
LF-DNN [34]	51.8	71.2	77.8	77.9	0.322	0.668
TFN [8]	53.3	70.9	78.1	78.1	0.322	0.662
LMF [35]	51.6	70.0	77.8	77.8	0.327	0.651
MFN [36]	55.4	72.7	79.4	79.4	0.301	0.712
Graph-MFN [1]	48.9	68.6	76.6	76.6	0.334	0.644
MuT [2]	54.6	74.2	80.8	80.7	0.300	0.738
MISA [9]	47.5	68.9	78.2	78.3	0.342	0.671
MAG-BERT [26]	49.2	70.6	77.1	77.1	0.346	0.641
Self-MM [37]	53.5	72.7	78.7	78.6	0.315	0.691
MMIM [3]	50.5	70.4	77.8	77.8	0.339	0.641
AV-MC [7]	52.1	73.2	80.6	80.7	0.301	0.721
KuDA [32]	53.1	74.3	80.2	80.1	0.289	0.741
KAN-MCP [5]	57.3	<u>75.0</u>	<u>81.6</u>	<u>81.7</u>	<u>0.281</u>	<u>0.742</u>
Ours	<u>57.1</u>	75.6	82.5	82.6	0.270	0.748

to 88.83 and alignment alone to 88.61 (disentanglement is the stronger regularizer); both together reach 89.18, and adding UAMG gives the best Acc-2 (89.62), F1 (89.61), MAE (0.6078) and Corr (0.8587)—every metric except 7-class accuracy, analyzed next. Crucially, the gate is computed in closed form from the IB posterior, reusing quantities the encoder already produces.

On 7-class accuracy. The only metric UAMG does not improve is Acc-7 (48.63 $\rightarrow$ 48.32, about two of the 685 test utterances). With six internal boundaries versus the single one of Acc-2, Acc-7 is the most sensitive to sub-bin perturbations: UAMG tightens the regression fit (MAE 0.6136 $\rightarrow$ 0.6078), but shrinkage

Table 3: Component ablation on CMU-MOSI. Align denotes cross-modal alignment, Disent the shared/specific disentanglement, and UAMG the uncertainty-aware modality gate.

Align	Disent	UAMG	Acc-7↑	Acc-2↑	F1↑	MAE↓	Corr↑
×	×	×	48.47	88.24	88.20	0.6264	0.8536
×	✓	×	48.71	88.83	88.80	0.6198	0.8552
✓	×	×	48.56	88.61	88.58	0.6212	0.8547
✓	✓	×	48.63	89.18	89.16	0.6136	0.8571
✓	✓	✓	48.32	89.62	89.61	0.6078	0.8587

Table 4: Comparison of modality-gating variants on CMU-MOSI. All gated variants derive the gate from the posterior standard deviation; we contrast letting the gradient flow into the covariance branch (grad. enabled) against detaching it (grad. detached).

Variant	Acc-7↑	Acc-2↑	F1↑	MAE↓	Corr↑
w/o gate	48.63	89.18	89.16	0.6136	0.8571
Grad. enabled	48.44	89.34	89.32	0.6115	0.8579
Grad. detached	48.32	89.62	89.61	0.6078	0.8587

toward the consensus can nudge a near-boundary sample into an adjacent bin. On the $6.8\times$ larger CMU-MOSEI test set the full model instead attains the best Acc-7 (54.68), so the dip reflects small-test-set sensitivity, not a deficiency.

Gating design. Table 4 compares three gating designs. Removing the gate is weakest (89.18); driving it from the posterior standard deviation while letting gradients flow into the covariance branch helps modestly (89.34); detaching that signal—so the posterior stays a clean uncertainty estimate—is best on Acc-2 (89.62), F1 (89.61), MAE (0.6078) and Corr (0.8587), at a marginal Acc-7 cost (48.63 $\rightarrow$ 48.32). We therefore adopt the detached variant.

5.3 Hyperparameter Sensitivity

Table 5 sweeps w_{disent} , w_{align} , and the gate temperature T_g one at a time; all three peak at the default. Acc-2 rises from 89.19 at $w_{\text{disent}}=0$ to 89.62 at 0.05 then declines, with w_{align} optimal at the same 0.05; either regularizer over-constrains the shared subspace when too large. T_g peaks at 2.0. Apart from the degenerate zero-weight cases, Acc-2 stays within a ~ 0.4 -point band (with F1, MAE, and Corr in Table 5 moving consistently), confirming stability.

5.4 Robustness Analysis

To probe inference-time robustness, we corrupt each modality of the CMU-MOSI test set by feature dropout (zeroing a fraction of dimensions) while keeping the other two intact, comparing the full model against an ablated variant without

Table 5: Hyperparameter sensitivity on CMU-MOSI. Each block varies one factor with the remaining settings fixed at the default configuration.

Factor	Value	Acc-2 $\uparrow$	F1 $\uparrow$	MAE $\downarrow$	Corr $\uparrow$
w_{disent}	0.00	89.19	89.17	0.6134	0.8572
	0.01	89.36	89.34	0.6109	0.8578
	0.05	89.62	89.61	0.6078	0.8587
	0.10	89.53	89.50	0.6087	0.8584
	0.20	89.29	89.27	0.6116	0.8576
w_{align}	0.00	89.26	89.23	0.6123	0.8574
	0.01	89.44	89.41	0.6099	0.8580
	0.05	89.62	89.61	0.6078	0.8587
	0.10	89.55	89.52	0.6085	0.8584
	0.20	89.31	89.28	0.6112	0.8575
T_g	0.5	89.38	89.35	0.6107	0.8579
	1.0	89.53	89.50	0.6088	0.8583
	2.0	89.62	89.61	0.6078	0.8587
	4.0	89.47	89.44	0.6097	0.8581

Table 6: Robustness under modality-specific feature dropout on CMU-MOSI. The listed modality is corrupted with dropout ratio p while the other two are kept intact.

Modality	Variant	p	Acc-2 $\uparrow$	F1 $\uparrow$	MAE $\downarrow$	Corr $\uparrow$
None	w/o UAMG	0.0	89.18	89.16	0.6136	0.8571
	Full model	0.0	89.62	89.61	0.6078	0.8587
Text	w/o UAMG	0.3	85.96	85.90	0.6912	0.8190
	Full model	0.3	87.34	87.29	0.6685	0.8317
	w/o UAMG	0.5	81.88	81.73	0.7824	0.7589
	Full model	0.5	84.52	84.41	0.7391	0.7864
Audio	w/o UAMG	0.3	88.44	88.39	0.6297	0.8464
	Full model	0.3	88.96	88.92	0.6209	0.8509
	w/o UAMG	0.5	87.31	87.24	0.6488	0.8361
	Full model	0.5	88.10	88.05	0.6356	0.8435
Visual	w/o UAMG	0.3	88.57	88.53	0.6268	0.8483
	Full model	0.3	89.03	89.00	0.6195	0.8520
	w/o UAMG	0.5	87.66	87.60	0.6432	0.8394
	Full model	0.5	88.35	88.31	0.6321	0.8457

UAMG (Table 6). The gate’s value grows sharply as the most informative modality degrades: under text dropout it leads the gate-free variant by 1.38 points at $p=0.3$ and 2.64 at $p=0.5$. Corrupting the weaker audio or visual streams causes far smaller drops, yet UAMG still recovers 0.5–0.8 points throughout, down-weighting a modality the moment the posterior flags it unreliable.

Table 7: End-to-end cost of the full model (CM-DRD-MIB + UAMG).

Metric	Value
Total parameters	184.30 M
Non-trainable parameters	284
Latency (bs = 16)	2.59 ms/sample
Throughput	~386 samples/s

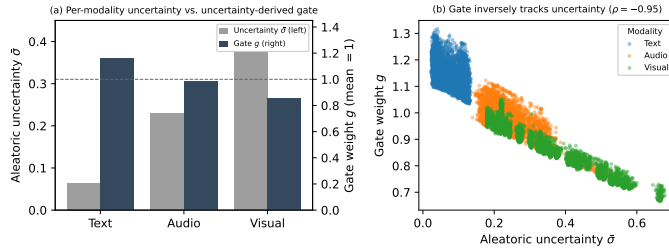


Fig. 2: UAMG analysis on the test set. (a) Mean aleatoric uncertainty $\bar{\sigma}$ (left axis) and the uncertainty-derived gate weight g (right axis, mean normalized to 1) per modality. (b) Per-sample gate weight versus uncertainty; the gate inversely tracks uncertainty ($\rho = -0.95$).

5.5 Efficiency Analysis

Table 7 reports end-to-end cost. The 184.30M parameter budget is dominated by the pretrained backbones; CM-DRD-MIB adds only compact variational μ/σ heads and a single-hidden-layer fusion KAN (4 neurons), and UAMG is a near parameter-free softmax over the negated posterior deviations. Inference runs at 2.59 ms per sample (batch size 16), about 386 samples/s on a single NVIDIA A800, so the gains stem from a more appropriate inductive bias rather than added capacity.

5.6 Representation Visualization

Uncertainty-aware gating. Figure 2 analyzes UAMG on the CMU-MOSEI test set. The three modalities carry markedly different aleatoric uncertainty (text most reliable at $\bar{\sigma} = 0.064$, then audio 0.229, visual 0.376), and the gate orders them inversely, tracking uncertainty per sample with a strong negative correlation ($\rho = -0.95$). The gate is sample-adaptive, not a fixed profile: per-sample weights span text [1.03, 1.32], audio [0.75, 1.12] and visual [0.67, 1.05]; text stays top-gated as the most reliable stream, yet the gate still reorders the weaker pair (audio outranks visual on 86.1% of samples), confirming genuine per-sample reweighting.

Shared/specific disentanglement. Figure 3 visualizes, via t-SNE, the CMU-MOSEI test embeddings under three views—fused, shared-only, and specific-only—colored by sentiment polarity. The fused and shared embeddings separate

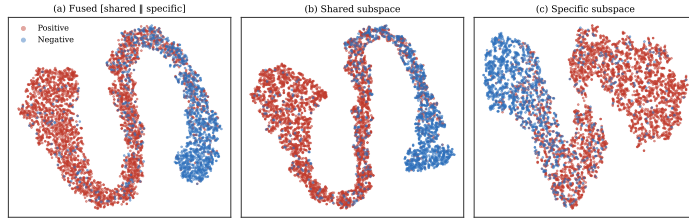


Fig. 3: T-SNE of test embeddings colored by sentiment polarity. (a) Fused [shared || specific]; (b) shared subspace; (c) specific subspace.

positive from negative clearly, whereas the specific subspace is less separable. Quantitatively, the binary-polarity silhouette score is graded—shared 0.49, fused 0.47, specific 0.42—so all three keep some polarity structure but to clearly different degrees, matching the intended split between cross-modal consensus and modality-private variation.

6 Conclusion

This paper presented DUKAN, a disentangled and uncertainty-aware framework on an interpretable KAN fusion backbone for multimodal sentiment analysis. Its CM-DRD-MIB encoder disentangles each modality into an InfoNCE-aligned shared subspace and a Barlow-decorrelated specific subspace, while the complementary UAMG module turns the per-modality posterior standard deviations into a per-sample modality gate; both train under Pareto-coordinated objectives. On CMU-MOSI, CMU-MOSEI, and the Chinese CH-SIMS v2, DUKAN attains state-of-the-art or competitive performance (Acc-2 89.62, F1 89.61 on CMU-MOSI; best Acc-7 54.68 on CMU-MOSEI) with improved robustness to modality corruption. Remaining challenges include utterance-level annotations that may oversimplify nuanced affect and the lack of longer-range conversational context; extending the design to these broader settings is future work.

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